

Data Analysis and Storytelling

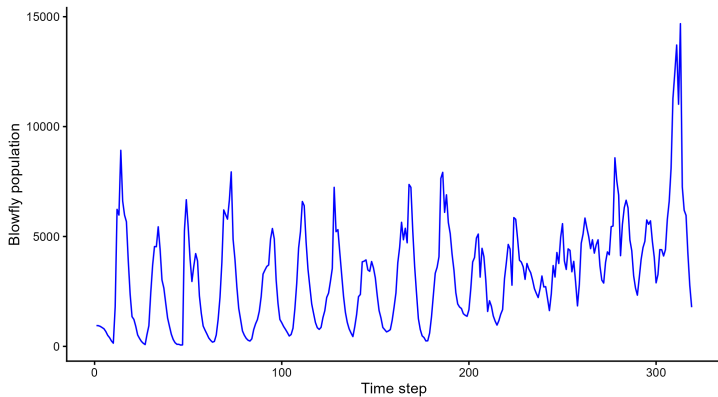
Regression analysis - model for time series

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Time series - a motivating example



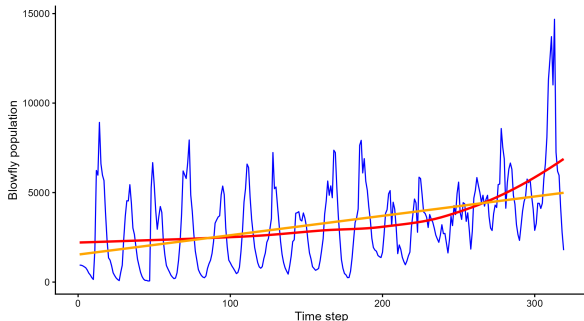
Different aims of adding to time series modelling

Adding a temporal component to a regression model can serve different aims:

- Estimate trends in the data
- Control for temporal autocorrelation

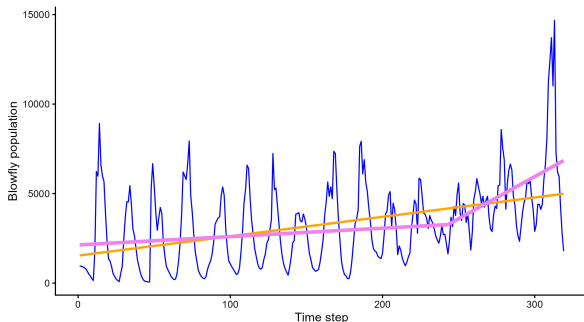
Trend analysis

- Trend analysis aims at estimating how the response variable change over time (blowfly number, tree mortality ...)
- Choosing the right functional form for the trend depend on both the complexity of the trend and the study aim - more complex form fit the data better at the cost of interpretability
- Specific models also exists to spot breakpoints in time series - system shifts



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Temporal autocorrelation

- Temporal autocorrelation is the fact that observations close together in time are more correlated than observation further apart
- Uncertainty estimates (from confidence intervals to p-values) from regression models depend on the independence between observations which can be violated by temporal autocorrelation
- The presence of temporal autocorrelation need to be tested on the residuals of the model, covariables may absorb some of the temporal signal - i.e. water stress has a strong effect on mortality and changes over time.

Steps to check temporal autocorrelation

- 1 Fit a regression model

```
m <- lm(mortality ~ stress)
```

- 2 check the residuals temporal autocorrelation

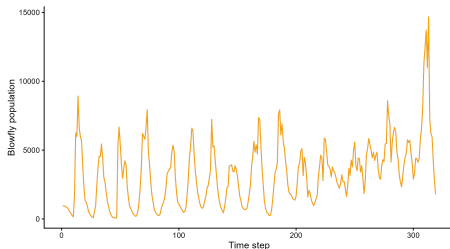
```
acf(residuals(m))
```

Expanding model to remove temporal autocorrelation

The hierarchical models we saw previously are a way to remove temporal autocorrelation:

$$\mu_{it} = \beta_0 + \gamma_t + \dots$$

$$\gamma_t \sim N(0, \sigma_{time})$$



Explicit temporal structure

- This model does not take into account the temporal proximity between observations
- An explicit model taking this into account is the autoregressive model:

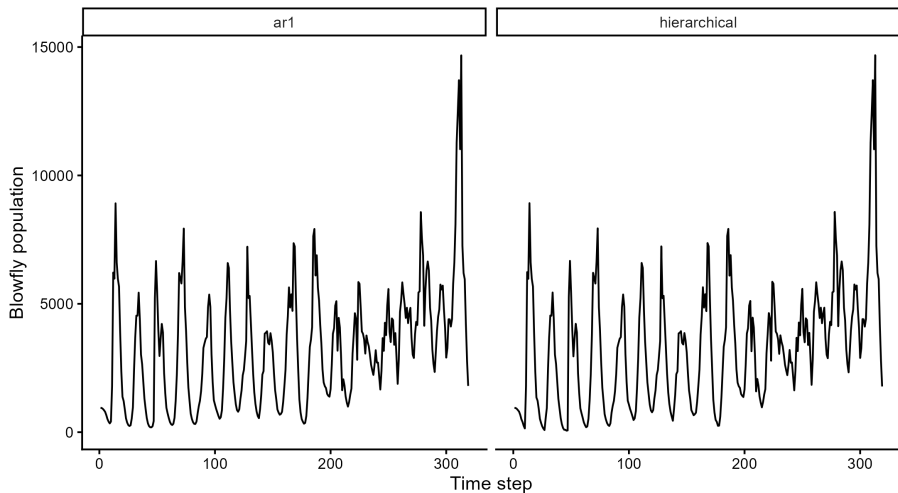
$$\mu_{it} = \beta_0 + \gamma_t + \dots$$

$$\gamma_t = \rho * \gamma_{t-1} + \delta_t$$

$$\delta_t \sim N(0, \sigma_{ar} * (1 - \rho^2)), \gamma_1 \sim N(0, \sigma_{ar})$$

- The AR1 parameters are: ρ the amount of autocorrelation varying from 0 (no autocorrelation) to 1 (complete autocorrelation) and σ_{ar} the standard deviation of the AR process, larger values meaning larger temporal noise.

Comparing AR1 and hierarchical models



AR1 vs hierarchical model

- AR1 models explicitly the temporal structure, an estimation of the temporal autocorrelation is given, which can be interpreted and compared
- The hierarchical model is lighter to implement and can erase just as good temporal autocorrelation signals

Combining trend and autocorrelation

Models can be developed to estimate both the trend and to account for autocorrelation via AR1 process, for instance we can add to a linear temporal trend a time varying intercept:

$$bowlfly_t \sim N(\mu_t, \sigma_{res})$$

$$\log(\mu_t) = \beta_0 + \gamma_t + \beta_1 * t$$

$$\gamma_t \sim N(0, \sigma_{hier})$$

Wrap up